

ANTIPAYUTA, Russian Federation

WMO: 230580

Lat: 69.104N

Lon: 76.858E

Elev: 3

StdP: 101.28

Time Zone: 5.00 (E05)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-43.4	-41.0	-45.9	0.0	-42.8	-44.0	0.1	-40.7	15.5	-9.9	13.8	-10.7	2.6	60	0.954

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.7	24.9	18.3	21.5	16.4	18.4	14.7	18.8	23.4	17.1	21.6	15.0	18.1	3.1	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.9	12.1	21.7	14.9	10.6	19.4	13.2	9.5	17.1	53.2	23.3	48.0	22.0	42.0	18.0	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.8	11.2	9.9	DB	-44.2	27.5	3.8	2.8	-46.9	29.5	-49.2	31.2	-51.3	32.8	-54.1	34.8	(4)
			WB	-43.7	19.6	2.8	2.4	-45.7	21.4	-47.4	22.8	-49.0	24.2	-51.0	25.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-8.4	-26.2	-25.0	-18.6	-13.5	-5.1	6.2	13.2	8.9	4.2	-5.8	-18.1	-22.2
	DBStd	15.22	8.69	9.85	9.19	8.01	4.94	5.21	5.12	3.16	3.60	7.04	8.19	9.14
	HDD10.0	6888	1121	980	886	704	467	139	22	61	176	491	844	998
	HDD18.3	9773	1379	1213	1144	954	725	363	177	293	424	749	1094	1257
	CDD10.0	176	0	0	0	0	0	26	122	26	3	0	0	0
	CDD18.3	18	0	0	0	0	0	0	18	0	0	0	0	0
	CDH23.3	149	0	0	0	0	0	1	148	0	0	0	0	0
	CDH26.7	31	0	0	0	0	0	0	31	0	0	0	0	0
Wind	WSAvg	4.8	4.8	4.9	5.0	5.2	5.6	4.7	4.5	4.5	4.3	4.5	4.6	4.8
Precipitation	PrecAvg	249	13	13	14	14	14	25	25	39	31	25	17	15
	PrecMax	507	32	34	43	44	31	139	133	90	64	51	49	46
	PrecMin	105	3	0	2	0	2	4	1	2	3	7	2	5
	PrecStd	93	7	10	10	10	9	26	27	27	16	11	10	10
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-3.9	-3.3	0.1	2.2	6.8	22.3	29.2	19.4	15.4	8.3	0.2	-2.4
		MCWB	-4.1	-3.8	-0.2	0.9	5.3	15.8	20.3	15.1	12.2	7.4	0.0	-2.7
	2%	DB	-6.2	-5.7	-1.7	1.1	3.5	19.6	27.3	17.4	12.9	5.2	-1.7	-4.3
		MCWB	-6.4	-5.9	-2.1	0.4	2.4	14.6	19.3	14.4	10.3	4.4	-1.9	-4.6
	5%	DB	-9.2	-7.7	-3.5	0.3	2.1	17.0	24.7	15.6	10.9	3.8	-3.5	-6.2
		MCWB	-9.5	-8.0	-3.9	-0.1	1.3	12.7	18.2	13.2	9.2	3.1	-3.7	-6.5
	10%	DB	-14.0	-10.2	-5.3	-0.9	1.2	14.2	21.9	14.2	9.5	2.1	-5.6	-8.5
		MCWB	-14.3	-10.6	-5.6	-1.3	0.6	11.0	17.0	12.2	8.3	1.5	-5.9	-8.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-4.2	-3.6	-0.2	1.2	5.2	16.9	22.0	16.4	12.9	7.3	0.0	-2.7
		MCDB	-4.1	-3.4	-0.1	1.6	6.6	21.1	27.8	18.5	14.7	8.0	0.1	-2.5
	2%	WB	-6.4	-5.9	-2.1	0.6	2.5	14.9	20.3	14.9	10.7	4.6	-1.9	-4.6
		MCDB	-6.1	-5.7	-1.8	1.1	3.3	18.8	25.4	16.8	12.5	5.1	-1.7	-4.4
	5%	WB	-9.4	-8.0	-3.9	-0.1	1.4	13.1	18.6	13.6	9.5	3.1	-3.7	-6.4
		MCDB	-9.1	-7.7	-3.6	0.3	1.9	16.5	23.2	15.2	10.8	3.7	-3.5	-6.2
	10%	WB	-14.2	-10.5	-5.6	-1.4	0.7	11.5	17.4	12.5	8.4	1.6	-5.9	-8.8
		MCDB	-14.0	-10.1	-5.3	-0.9	1.2	14.0	21.9	13.9	9.4	2.0	-5.5	-8.5
Mean Daily Temperature Range	5% DB	MDBR	7.0	7.7	8.5	9.2	5.6	6.1	7.7	5.9	4.8	5.2	7.1	8.1
		MCDBR	10.4	8.9	9.1	6.8	4.5	10.6	10.7	7.9	7.1	3.7	7.4	8.8
		MCWBR	10.4	8.9	9.0	6.7	3.8	6.5	4.6	5.3	5.1	3.3	7.3	8.6
	5% WB	MCDBR	10.3	9.0	9.0	7.1	4.1	10.0	10.3	7.1	6.3	3.4	7.2	8.8
		MCWBR	10.3	8.9	9.1	7.1	3.6	6.7	5.2	5.4	4.9	3.1	7.2	8.6
Clear-Sky Solar Irradiance	taub	0.209	0.248	0.301	0.321	0.339	0.359	0.386	0.369	0.317	0.252	0.198	N/A	
	taud	1.867	1.981	2.024	2.066	2.124	2.345	2.329	2.356	2.429	2.151	1.901	N/A	
	Ebn at Noon	27	482	688	803	837	834	791	760	711	488	24	0	
	Edh at Noon	5	47	89	116	124	104	102	89	62	38	4	0	
All-Sky Solar Radiation	RadAvg	0.01	0.32	1.80	3.95	5.42	5.29	4.99	3.20	1.61	0.50	0.06	0.00	
	RadStd	0.00	0.06	0.20	0.35	0.32	0.44	0.64	0.43	0.18	0.07	0.01	0.00	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	+1.54	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page